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Group Assignment 1

Complete the following worksheet **as a group** and submit it to Canvas as a PDF. Only one person needs to submit the worksheet. Contributions should be approximately equal from each group member. There are three parts to complete.

Total points: 10

Part 1: Your Project

A) Describe your research question (2 sentences max). Be as specific as reasonably possible *(type of samples, key variables of interest, disease/environment being studied, any other key info).* **(1 point)**

Among 12 month old breastfed infants, how does iron supplementation type (MNP vs. FeSO4 vs. none) influence gut microbiome composition, and how do these microbial differences relate to hemoglobin and ferritin levels?

B) List and briefly define the key terms and parameters that are necessary to understand your project. Use the example table below and add more rows as necessary. You do not need to include microbiome terms that are introduced in class modules (ex. '16S'). **(1 point)**
For example: if studying whether the ***microbiome correlates with depression in people with Parkinson's disease****, you would need to define a)* ***Parkinson's disease*** *and b)* ***the metric that was used to describe depression in your dataset.*** *Include key info such as the name of the metric, categorical or numerical, what numerical cutoff you will use to separate depressed/not depressed participants, etc.*

Term	Definition/Description
FSS (Fatigue Severity Survey)	*Numeric variable. Scored out of 7. Higher numbers indicate more fatigue. We will simply divide people by scores of <4 and 4+ as there is no standard cutoff in the literature.*
Anemia	*Anemia is a deficiency of healthy red blood cells and hemoglobin to carry oxygen to tissues*
IDA (iron-deficiency anemia)	*Low levels of iron leading to reduced red blood cell production* - *Can cause fatigue, pale skin and dizziness*
MNP Supplement	*Multiple micronutrient powder: a combination of a handful of vitamins (A, C, D, E B1, B2, B3, B6, B12), iron, zinc, folic acid, copper, selenium, and iodine. Used to prevent malnutrition in children. Target age of MNP is generally between 6 to 59 months (target group: 6-24 months)*

	https://www.unicef.org/supply/media/16661/file/S1580201-Multiple-Micronutrient-powder-MNP-15-components-Specification.pdf
$FeSO_4$ Supplement	*Iron (II) sulfate or ferrous sulfate: widely used iron supplement to treat and prevent IDA. Can be found in tablet form or liquid form and most groups of people with IDA will be able to take the supplement.*
Ferritin	*A protein that stores iron and releases the iron in a controlled fashion. Low ferritin indicates low iron reserves in the body which could indicate IDA. Normal levels of ferritin in children between 6 months to 15 years is 7-140 ng/ml (i cant find a smaller range thats just for 6-12 month olds)*

C) Compile a concise list of all papers that have been published using this dataset (including the original dataset, UJEMI/UJEMI+, broader literature). Use the example table below and add more rows as necessary. **You only need to summarize the findings if it is related to your research focus** *(for example: if you are studying the effect of sleep on the Parkinson's disease microbiome, you don't need to summarize papers that focus on diet).* **(3 points)**

Type	Title	Focus	Summarized Findings
Original	*Anemia in infancy is associated with alterations in systemic metabolism and microbial structure and function in a sex-specific manner: an observational study*	*Anemia and microbiome in infants*	*Anemia patients have no structural or functional differences in their microbiome compared to healthy infants, but iron-deficiency anemia (IDA) did affect the microbiome (IDA patients had higher butyrate-producing bacteria than nonanemic infants). Additionally, there are differences in the metabolic results between males and females.*
UJEMI	*A breast milk exclusive diet promotes dysbiosis in the gut microbiome of six-month-old anemic infants*	*Breast milk diet and infant microbiome*	*Anemia status (anemic vs healthy) does not affect the microbiome diversity, but diet does. Infants with breast milk exclusive diets were found to have increased microbiome diversity compared to complete diet infants. However, this increased microbial diversity due to the breast milk diet is only found in anemic infants. Healthy*

			infants on breast milk diets had no increase in microbial diversity.
UJEMI	*Iron-deficiency anemia is associated with loss of gut microbial diversity in 6-and 12-month-old infants*	*IDA and microbial diversity in infants*	*6-and 12-month-old infants had lower microbial diversity than healthy infants. Furthermore, there are significant differences in the microbial communities between 6-and 12-month-old infants (12-month-old infants had a higher microbial diversity than 6-month-old infants).*
UJEMI	*Ironing Out Anemia: High Inflammation Modulates Microbial Composition and Indicates the Activation of Cellular Stress Responses in Infected Anemic Infants*	*Inflammation in anemic infants and the microbiome and cellular responses*	*This paper focuses on inflammation, while our focus is on anemia supplements. Therefore, we did not provide a summary for this paper because it is not related to our research focus.*
UJEMI	*Nutrient supplementation does not exacerbate pathogenic inflammation in infants with iron deficiency anemia*	*Anemia supplements' effects on inflammation in IDA infants*	*The stage of infection in anemic infants does not change their microbial diversity. Additionally, when analyzing only the effect of iron supplements on 12-month-old anemic infants, iron supplements were found to have no effect on microbial diversity. However, when also considering infection status, infants taking iron supplements have increased microbial diversity.*
UJEMI	*Anemia May Mask the Effects that Stunting has on the Dysbiosis of the Infant Gut Microbiota*	*Anemia, stunting, and the infant gut microbiome*	*This paper focuses on the relationship between anemia and stunting, and how they affect the microbial diversity of infants. Our research focus is on anemia supplements, so the summary for this paper is not included.*
UJEMI	*Non-breast milk diet increases gut microbial*	*Non-breast milk diets,*	*There is a difference in microbial diversity between 6-month-old*

	diversity and inflammation in six-month-old infants	*microbiome, and inflammation*	*infants on breast milk diets compared to non-breast milk diets. Non-breast milk diet infants have a much greater microbial diversity than infants with breast milk diets. Furthermore, there was found to be no significant difference in the inflammatory levels and microbial diversity of anemic and non-anemic infants with breast milk and non-breast milk diets.*
Literature	*Gut microbial ecology and function of a Pakistani cohort with Iron deficiency Anemia*	*Relationship between iron deficiency anemia, gut microbial diversity, and microbial metabolites*	*Sequencing meconium samples from infants born to healthy and anemic mothers the study found that iron deficiency anemia was associated with altered gut microbial composition and metabolic output. No statistical difference observed between groups in Shannon Diversity. However, enrichment of anaerobic, branched short-chain fatty acid producing genera alongside acetate and propionate production was linked to anemia. Findings suggest that through changes in microbial fermentation and metabolite profiles that iron metabolism can be influenced by anemia-associated dysbiosis.*
Literature	*Fermented Goat's Milk Contributes to the Recovery of Iron Deficiency Anemia via Modulation of the Gut Microbiome*	*Effect of dietary intervention on gut microbiome composition and diversity during iron deficiency anemia recovery*	*Study investigated the influence in gut microbiome when fermented goat's milk was given to iron deficient anemia rat models. Shannon diversity did not show a consistent difference between treatment groups. There was an increased microbial richness and shift in community composition*

			observed in iron-deficient rats that received fermented goat's milk. An association between microbiome changes were made with improved iron status which suggests that dietary modulation of gut microbial composition may support recovery from iron deficiency anemia without altering overall alpha diversity.

https://pubmed.ncbi.nlm.nih.gov/?linkname=pubmed_pubmed_citedin&from_uid=30351381
(for papers that cited original)

Part 2: Your Dataset

Run the QIIME2 pipeline on your final project dataset and answer the following questions. You do not need to generate any diversity metrics or functional annotations – just run the pipeline up to and including the rarefaction curve.
Save all the commands you use in a text file – you don't need to submit it with this assignment, but it is your record of the steps you performed, and you will publish it along with your manuscript.

A) Dataset folder name in project_2:

anemia

B) What denoising tool did you use?

dada2

C) Fill in the following table: **(1 point)**

	Before denoising	After denoising
Number of samples	193	193
Sequencing depth range (lowest-highest)	67-74,453	43-64,241

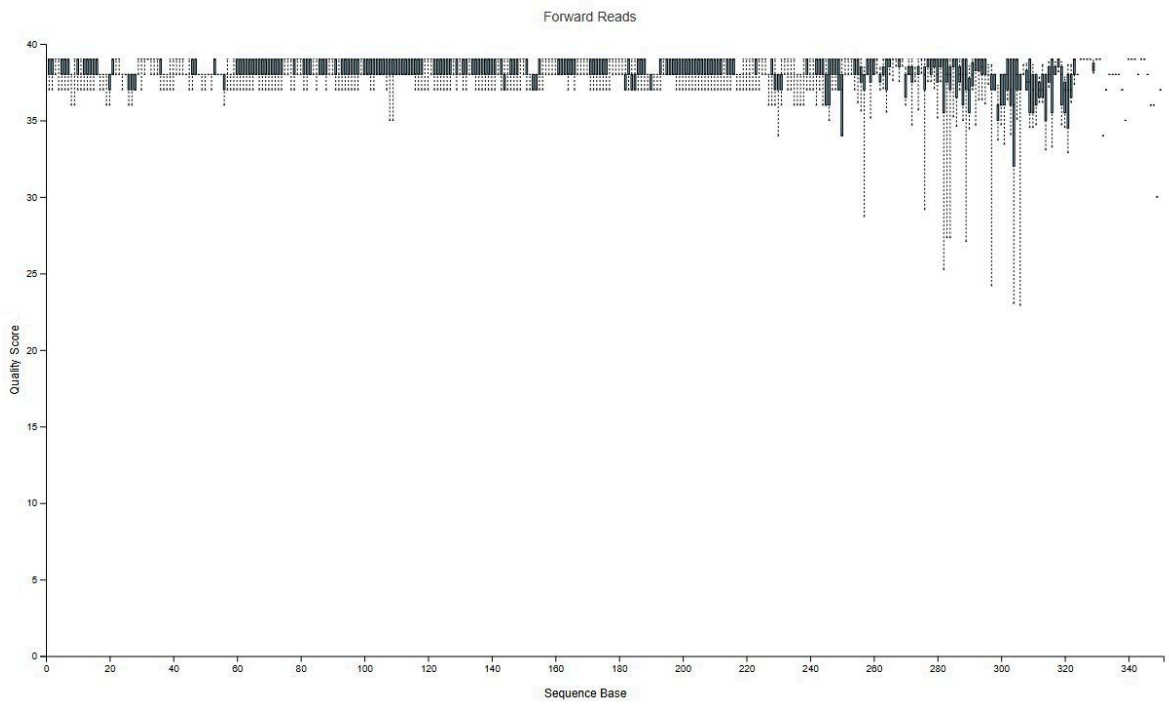
D) What is the maximum read length in the dataset? Do >99% your samples have the same read length? (yes/no). If not, briefly describe the distribution of read lengths. **(1 point)**

Max read length = 350 nucleotides. Not every sample hast the same length, there are some reads with 252 reads, some with 253 reads, and very few (< 2% with more than 253 reads).

E) What will you truncate your reads to? Paste your read quality plot below to justify your

truncation parameter. *Note: if all your reads are not the same length, you might want to try a few options and see what happens in your stats-dada2.qzv file.* **(1 point)**

We will be truncating these reads to 252 ntds long. The quality of the reads is not an issue, as seen below in the quality plot below. The median PHRED score of each nucleotide position does not drop below a score of 30. Therefore, we will be keeping as much information as possible while keeping the reads the same length (ie. 252 nucleotides long).



F) What rarefaction depth will you choose for your diversity analyses? Justify your answer by using a rarefaction curve and your table.qzv file. How many samples are discarded at this rarefaction depth? **(1 point)**

The rarefaction depth that we have chosen for our diversity analyses is 22,500. This is because a depth of 22,500 is comfortably past where the graph starts to plateau, as it begins to plateau at a sequencing depth of 5000. Additionally, the number of samples starts to go significantly down only after 22,500. Thus, a sampling depth of 22,500 will allow for a good number of samples to be retained while maintaining a good amount of biological variance.

Part 3: Your Groupwork (1 pt)

This is your first group assignment. After finishing Part 1 and 2, please take a few minutes as a group to reflect on your group's dynamics so far.

- Was it easy or difficult to divide the work equally?

I believe it was fairly easy. The workload was enough that we were able to divide it evenly among 5 people, and not one person was doing significantly more work than others.

- Were there any challenges, groupwork related or otherwise?

There were some challenges in terms of communication, as we were still getting used to how to coordinate with each other either to do code or to distribute any research work. However, in terms of coding, having a log of the commands we used when running QIIME2 on our dataset helped a lot especially when it came to tracking different files. In terms of research work, it was challenging to get a solid research question down, as we realized after sampling that certain groups just did not have enough samples for us to do a thorough analysis. Furthermore, it was a bit difficult to reason which articles that cite our main research article would be viable to include in part c, but eventually we were able to get over that hurdle.

- If applicable, how can things be improved for your next group assignment?

I believe having scheduled weekly meetings outside of our meetings with the TA would help, as there are times where we need to discuss our approach to the project as a group. Additionally, making internal deadlines for different parts of the project would help as well, so that we can make sure we are complete with everything once the actual due date comes around.